Step 6 : Critical value

Since H_1 is two-sided alternative hypothesis, the critical value at $\alpha = 0.01$ is

$t_e = t_{m+n-2, \frac{\alpha}{2}} = t_{16,0.005} = 2.921$

Step 7 : Decision

Since it is two-tailed test, elements of critical region are defined by the rejection rule $|t_0| < t_e = t_{m+n-2, \frac{\alpha}{2}} = t_{16,0.005} = 2.921$. For the given sample information $|t_0| = 1.3957 < t_e = 2.921$. It indicates that given sample contains insufficient evidence to reject H_0. Hence, the mean performance of the students in these batches are equal.

Example 2.4

Two types of batteries are tested for their length of life (in hours). The following data is the summary descriptive statistics.

Type	Number of batteries	Average life (in hours)	Sample standard deviation
A	14	94	16
B	13	86	20

Is there any significant difference between the average life of the two batteries at 5% level of significance?

Solution:

Step 1 : Hypotheses

Null Hypothesis $H_0 : \mu_X = \mu_Y$

i.e., there is no significant difference in average life of two types of batteries *A* and *B*.

Alternative Hypothesis $H_0 : \mu_X \neq \mu_Y$

i.e., there is significant difference in average life of two types of batteries *A* and *B*. It is a two-sided alternative hypothesis

Step 2 : Data

The given sample information are :

m = number of batteries under type A = 14

n = number of batteries under type B = 13

$\bar{x}$ = Average life (in hours) of type A battery = 94

$\bar{y}$ = Average life (in hours) of type B battery = 86

s_X = standard deviation of type A battery =16

s_Y = standard deviation of type B battery = 20

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